

Evidencias

02_simple_reflex_agent.py

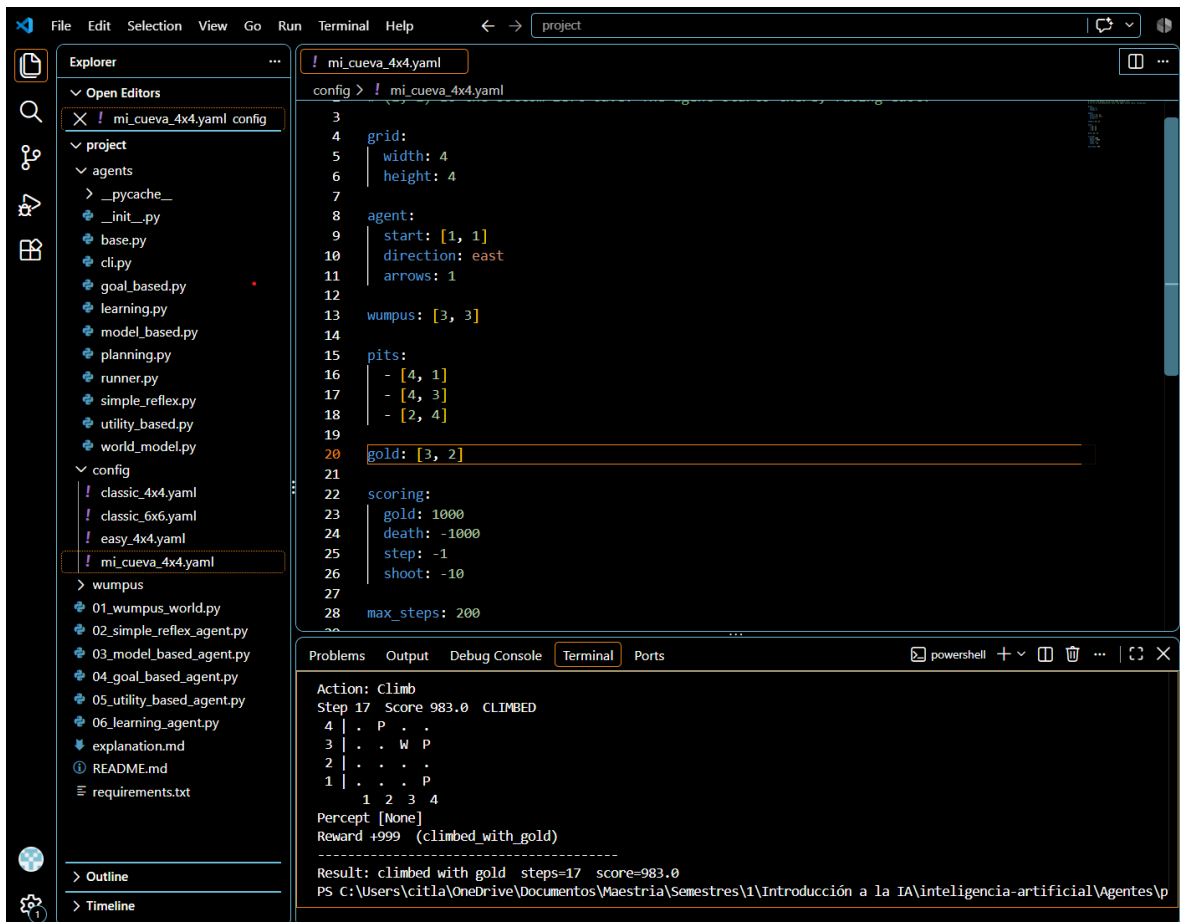
The image shows a VS Code editor window with a project named 'project'. The Explorer sidebar on the left shows the file structure, with 'mi_cueva_4x4.yaml' selected under the 'config' folder. The main editor displays the contents of this file, which is a YAML configuration for a game environment. The configuration includes a 4x4 grid, an agent starting at [1, 1] facing east, a wumpus at [3, 3], pits at [4, 1], [4, 3], and [2, 4], and gold at [3, 2]. Scoring is set to 1000 for gold, -1000 for death, -1 for steps, and -10 for shooting. The maximum steps are 200.

```
3
4 grid:
5   width: 4
6   height: 4
7
8 agent:
9   start: [1, 1]
10  direction: east
11  arrows: 1
12
13 wumpus: [3, 3]
14
15 pits:
16   - [4, 1]
17   - [4, 3]
18   - [2, 4]
19
20 gold: [3, 2]
21
22 scoring:
23   gold: 1000
24   death: -1000
25   step: -1
26   shoot: -10
27
28 max_steps: 200
```

The bottom panel shows the Terminal output, which displays the action taken (TurnRight), the current step (200), score (-200.0), and the current state (IN CAVE). It also shows a 4x4 grid representing the environment state, with 'P' for pits, 'W' for wumpus, 'G' for gold, and 'P' for the player. The percept is [Breeze], the reward is -1 (max steps), and the final result is 'stopped without gold steps=200 score=-200.0'.

```
-----
Action: TurnRight
Step 200  Score -200.0  IN CAVE
4 | .  P  .  .
3 | .  .  W  P
2 | .  .  G  .
1 | .  .  <  P
  | 1  2  3  4
Percept [Breeze]
Reward -1 (max steps)
-----
Result: stopped without gold  steps=200  score=-200.0
```

03_model_based_agent.py



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left displays the project structure, with the file `mi_cueva_4x4.yaml` selected under the `config` folder. The main editor displays the content of `mi_cueva_4x4.yaml`, which defines a 4x4 grid world environment. The terminal at the bottom shows the output of the program, indicating that the agent has successfully climbed with gold, achieving a score of 983.0 after 17 steps.

```
! mi_cueva_4x4.yaml
config > ! mi_cueva_4x4.yaml

3
4  grid:
5    width: 4
6    height: 4
7
8  agent:
9    start: [1, 1]
10   direction: east
11   arrows: 1
12
13  wumpus: [3, 3]
14
15  pits:
16    - [4, 1]
17    - [4, 3]
18    - [2, 4]
19
20  gold: [3, 2]
21
22  scoring:
23    gold: 1000
24    death: -1000
25    step: -1
26    shoot: -10
27
28  max_steps: 200

Problems Output Debug Console Terminal Ports
Action: Climb
Step 17 Score 983.0 CLIMBED
4 | . P . .
3 | . . W P
2 | . . . .
1 | . . . P
  1 2 3 4
Percept [None]
Reward +999 (climbed_with_gold)
-----
Result: climbed with gold steps=17 score=983.0
PS C:\Users\citla\OneDrive\Documentos\Maestria\Semestres\1\Introducción a la IA\inteligencia-artificial\Agentes\p
```

```
PS C:\Users\citla\OneDrive\Documentos\Maestria\Semestres\1\Introducción a la IA\inteligencia-artificial\Agentes\p
roject> python 03_model_based_agent.py --config .\config\mi_cueva_4x4.yaml
Step 0 Score 0 IN CAVE
4 | . P . .
3 | . . W P
2 | . . G .
1 | > . . P
  1 2 3 4
Percept [None]
Agent: model-based
-----
Action: TurnLeft
Step 1 Score -1.0 IN CAVE
4 | . P . .
3 | . . W P
2 | . . G .
1 | ^ . . P
  1 2 3 4
Percept [None]
Reward -1 (ok)
-----
Action: Forward
Step 2 Score -2.0 IN CAVE
4 | . P . .
3 | . . W P
2 | ^ . G .
```

04_goal_based_agent.py

```
Result: climbed with gold  steps=17  score=983.0
PS C:\Users\citla\OneDrive\Documentos\Maestria\Semestres\1\Introducción a la IA\inteligencia-artificial\Agentes\p
roject> python 04_goal_based_agent.py --config .\config\mi_cueva_4x4.yaml
Step 0  Score 0  IN CAVE
  4 | .  P  .  .
  3 | .  .  W  P
  2 | .  .  G  .
  1 | > .  .  P
    1 2 3 4
Percept [None]
Agent: goal-based
-----
Action: TurnLeft
Step 1  Score -1.0  IN CAVE
  4 | .  P  .  .
  3 | .  .  W  P
  2 | .  .  G  .
  1 | ^ .  .  P
    1 2 3 4
Percept [None]
Reward -1  (ok)
-----
Action: Forward
Step 2  Score -2.0  IN CAVE
  4 | .  P  .  .
  3 | .  .  W  P
```

```
  1 | .  .  .  P
    1 2 3 4
Percept [None]
Reward -1  (ok)
-----
Action: Forward
Step 16  Score -16.0  IN CAVE
  4 | .  P  .  .
  3 | .  .  W  P
  2 | .  .  .  .
  1 | v .  .  P
    1 2 3 4
Percept [None]
Reward -1  (moved)
-----
Action: Climb
Step 17  Score 983.0  CLIMBED
  4 | .  P  .  .
  3 | .  .  W  P
  2 | .  .  .  .
  1 | .  .  .  P
    1 2 3 4
Percept [None]
Reward +999  (climbed_with_gold)
-----
Result: climbed with gold  steps=17  score=983.0
```

05_utility_based_agent.py

```
Problems Output Debug Console Terminal Ports powershell + v [ ] [ ] ... | [ ] [ ] X
PS C:\Users\citla\OneDrive\Documentos\Maestria\Semestres\1\Introducción a la IA\inteligencia-artificial\Agentes\p
roject> python 05_utility_based_agent.py --config .\config\mi_cueva_4x4.yaml
Step 0 Score 0 IN CAVE
  4 | . P . .
  3 | . . W P
  2 | . . G .
  1 | > . . P
    1 2 3 4
Percept [None]
Agent: utility-based
-----
Action: Forward
Step 1 Score -1.0 IN CAVE
  4 | . P . .
  3 | . . W P
  2 | . . G .
  1 | > . P
    1 2 3 4
Percept [None]
Reward -1 (moved)
-----
Action: Forward
Step 2 Score -2.0 IN CAVE
  4 | . P . .
  3 | . . W P
  2 | . . G .
```

```
Problems Output Debug Console Terminal Ports powershell + v [ ] [ ] ... | [ ] [ ] X
  1 | . . . P
    1 2 3 4
Percept [None]
Reward -1 (ok)
-----
Action: Forward
Step 33 Score -43.0 IN CAVE
  4 | . P . .
  3 | . . W P
  2 | . . . .
  1 | v . . P
    1 2 3 4
Percept [None]
Reward -1 (moved)
-----
Action: Climb
Step 34 Score 956.0 CLIMBED
  4 | . P . .
  3 | . . W P
  2 | . . . .
  1 | . . . P
    1 2 3 4
Percept [None]
Reward +999 (climbed_with_gold)
-----
Result: climbed with gold steps=34 score=956.0
```

06_learning_agent.py

```
Problems Output Debug Console Terminal Ports powershell + - [ ] [X] ... | [ ] [X] X
roject> python 06_learning_agent.py --config .\config\mi_cueva_4x4.yaml
Trained 1500 episodes on config\mi_cueva_4x4.yaml
Mean score (all): 463.7
Mean score (last 50): 702.5
Q-table size: 690 state-action pairs

--- Greedy demo (epsilon = 0) ---

Step 0 Score 0 IN CAVE
 4 | . P . .
 3 | . . W P
 2 | . . G .
 1 | > . . P
    1 2 3 4
Percept [None]
Agent: learning
-----
Action: Forward
Step 1 Score -1.0 IN CAVE
 4 | . P . .
 3 | . . W P
 2 | . . G .
 1 | . > . P
    1 2 3 4
Percept [None]
Reward -1 (moved)
```

```
Problems Output Debug Console Terminal Ports powershell + - [ ] [X] ... | [ ] [X] X
Percept [None]
Reward -1 (moved)
-----
Action: Forward
Step 11 Score -11.0 IN CAVE
 4 | . P . .
 3 | . . W P
 2 | . . . .
 1 | < . . P
    1 2 3 4
Percept [None]
Reward -1 (moved)
-----
Action: Climb
Step 12 Score 988.0 CLIMBED
 4 | . P . .
 3 | . . W P
 2 | . . . .
 1 | . . . P
    1 2 3 4
Percept [None]
Reward +999 (climbed_with_gold)
-----
Result: climbed with gold steps=12 score=988.0
```